

Project Matter-Laser: Integrated System Specification

HARDWARE REGISTRY, MATERIAL INTERFACES, & PERFORMANCE SPECS (PHASE V ON-CHIP VARIANT)

Document ID: ML-SYS-REG-2026

Status: Validated System Blueprint

1. System Architecture Overview

This configuration details an on-chip, solid-state variant of the Matter-Laser project. It uses a synchronized phased array of $N=8$ semiconductor injectors to eliminate phase slippage, generating a continuous, highly stable, non-ionizing 100 THz superradiant output while managing thermal strain boundaries down to 4.2 Kelvin.

2. Hardware Registry & Layer Stack Specification

Component / Layer	Material Selection	Physical Dimensions	Functional Mandate & Control Mechanics
Base Support Carrier	Synthetic CVD Diamond (Alternative: Sapphire Al ₂ O ₃)	Thickness: 500 μm Footprint: 10 mm × 10 mm	Acts as a master thermal sink to instantly dissipate localized heating into the cryo-reservoir. Clamps lateral thermal mechanical expansion.
Quantum Emitting Array	Strained GaAsP Superlattice Matrix (N = 8 Elements)	Array Element Pitch: 200 nm Local Beamlet Spread: 10 nm	Simultaneous optical injection from a single split femtosecond seed laser source. Drives ballistic emission of 71.1 eV electron packets in-phase.
Interfacial Strain Buffer	Sputtered Titanium-Tungsten (TiW) Graded Alloy	Thickness: 2.0 nanometers	Acts as a chemical and mechanical shock absorber at the junction boundary. Absorbs the expansion mismatch between the substrate and the gold mesh.
Resonant Extraction Mesh	Pure Gold (Au) Film Matrix	Period (L _g): 50 nm Tooth Width: 25 nm Slot Depth: 15 nm	Linear profile converts the passing matter-wave into a 100 THz output via Smith-Purcell interaction. Features outer perimeter Zig-Zag spring anchors.
Vacuum Clearance Gap	Ultra-High Vacuum (UHV) Space Envelope	Clearance Height (h _c): 5 to 8 nm Pressure: ≤ 10 ⁻¹⁰ Torr	Allows ballistic fly-over of the merged electron wavefront without physical collision or surface scattering.
Signal Collector	Lithographic Tapered Metallic Vivaldi Horn	Length: 1.5 μm	Captures the perpendicular 100 THz electromagnetic wave and smoothly routes it 90 degrees into the horizontal circuit plane.
Logic Routing Layer	Silicon-on-Insulator (SOI) + Atomically Thin Graphene Channel	Channel Width: 40 nm Monolayer Lattice Matrix	Conveys the 100 THz energy as Surface Plasmon Polaritons (SPPs). Serves as a femtosecond electro-optical logic gate (0 or 1).

3. Output Performance Specifications

Performance Metric	Calculated Design Metric	Operational Domain Status
Primary Target Frequency	100.0 Terahertz (THz)	Far-Infrared Spectrum
Photon Energy Output	0.413 Electron-Volts (eV)	Non-Ionizing Radiation
Matter-Wave de Broglie Scale	0.145 nanometers (145 pm)	Sub-Atomic Scale
Net Power Scaling Mode	Superradiant Continuous Wave ($P \propto N^2$)	7.4x Baseline Amplified
Spatial Emission Profile	$\pm 5.0^\circ$ Symmetrical Forward Cone	Hyper-Collimated Beam
Spectral Linewidth	$\Delta \nu / \nu \leq 0.01\%$	High Optical Coherence

4. Multi-Scale Functional Applications

- **Petabit Signal Computing:** Enables clock switching speeds orders of magnitude faster than silicon-gate transistors, serving as a platform for ultra-fast, non-thermal data processors.
- **Non-Destructive Material Tomography:** 0.41 eV photons penetrate materials safely without causing atomic bond ionization, enabling real-time structural inspection of micro-components.
- **On-Chip Spectroscopy:** High-coherence 100 THz light allows for immediate vibrational and rotational identification of unknown organic molecules or trace chemical compounds.